

CURVED 3-SEAT SOFA / CNC FRAME

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF

1 - SPECIFICATION, MATERIAL AND BILL OF MATERIALS

SPECIFICATION

Overall width (chord of the back)	2200 mm
Overall depth over the crescent	1074 mm
Radial depth of the seating band	900 mm
Plan sweep / back radius	60 deg / R2200 mm
Usable seat arc between the arms	1354 mm (3 x 451)
Plinth height	45 mm (3 laminae)
Seat frame above the floor	345 mm
Seat height with a 100 mm cushion	~445 mm
Arm height above the floor	620 mm
Backrest height above the floor	760 mm
Joint clearance	slots 15.4 mm for 15 mm plate
Corner relief	dogbone R3.2 mm (6 mm cutter)

MATERIAL CONSUMPTION

MATERIAL	SHEETS	PIECES	NET AREA	SHEET USE	CUT PATH
15 mm PLYWOOD (2440 x 1220)	7	37	6.42 m2	31 %	126 m
4 mm FLEXIBLE PLYWOOD (2440 x 1220)	1	3	2.01 m2	68 %	13 m

TOTAL 8 sheets - bare frame mass ~63 kg at 600 kg/m3, before foam and fabric

Buy one spare 15 mm sheet: the nesting is shelf-packed, and a mis-cut curved rail is the expensive one to redo.

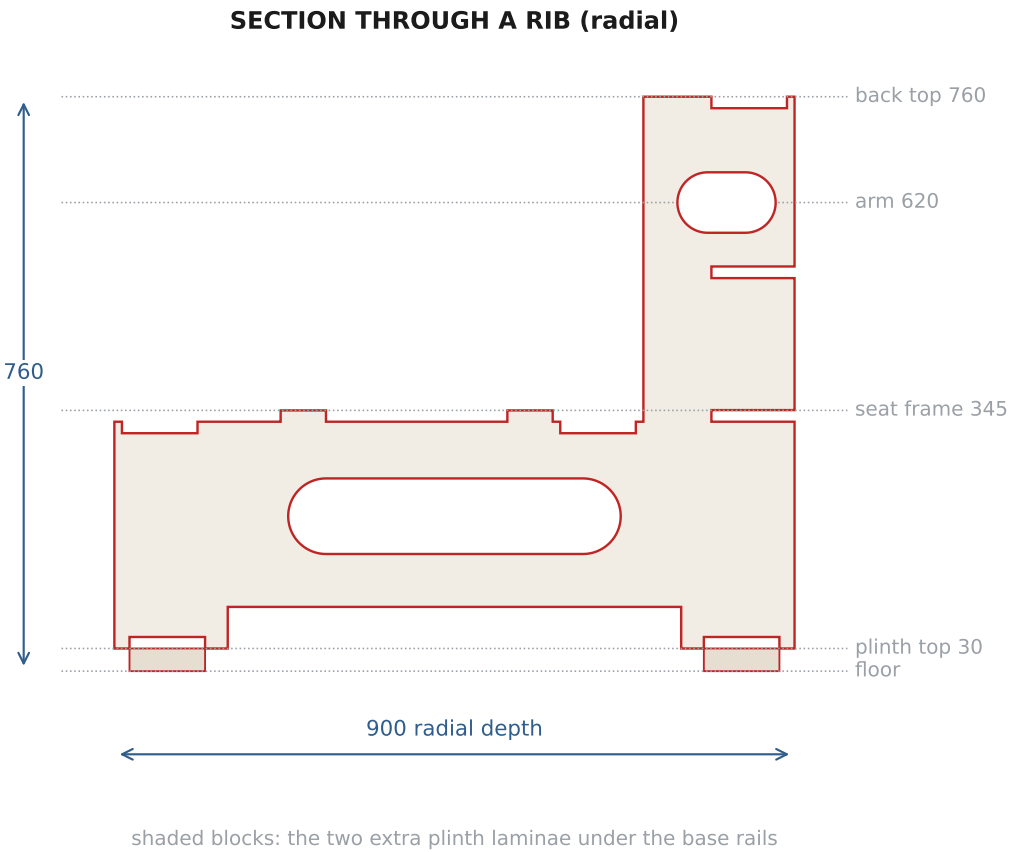
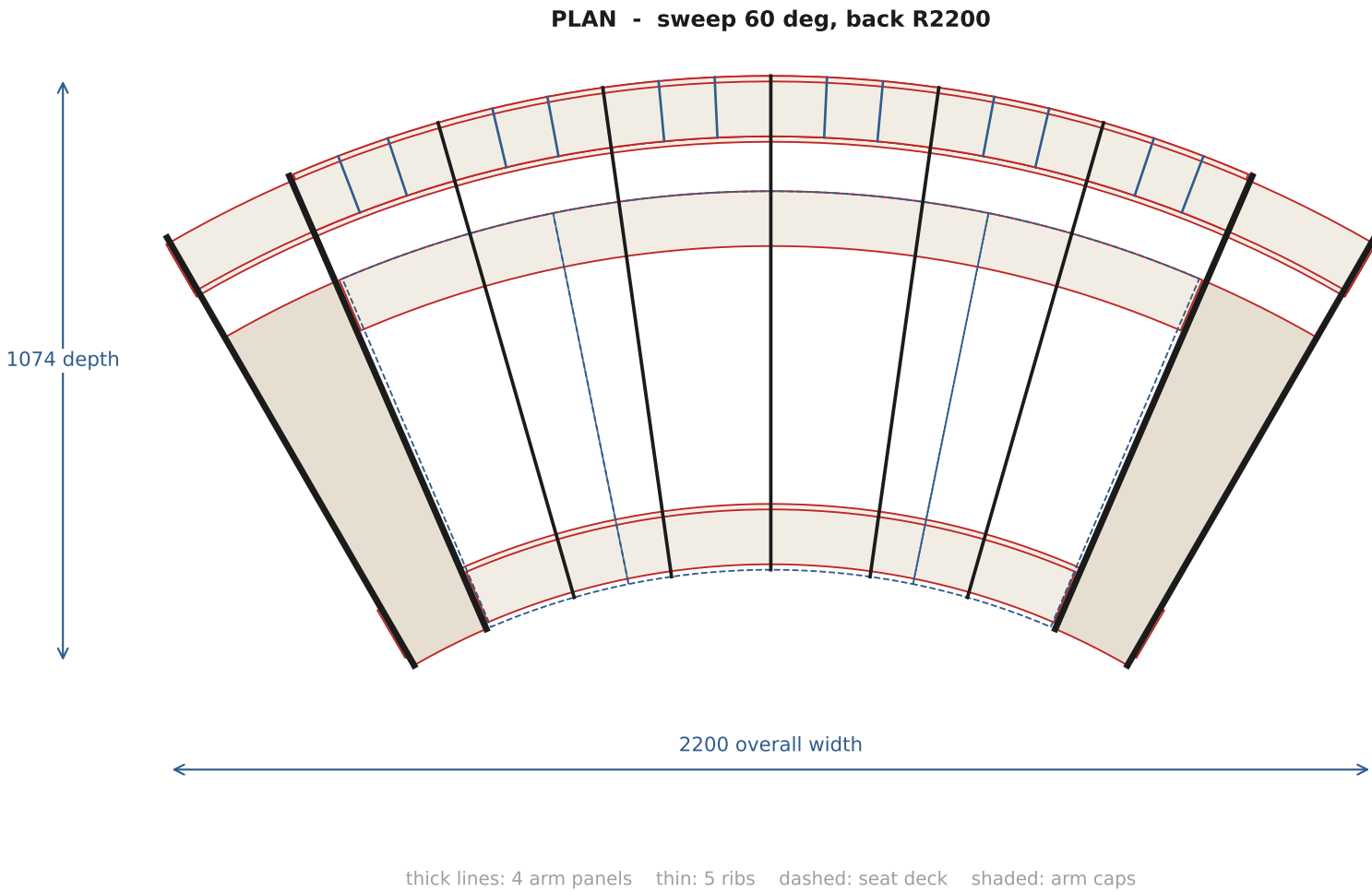
ASSEMBLY SEQUENCE

- Glue up the plinth: 3 laminae each of RAIL-BASE-IN and RAIL-BASE-OUT.
- Drop the 5 RIBs and the 2 ARM-PANEL-INS onto the plinth rails.
- Drop RAIL-SEAT-IN / -OUT and RAIL-BACK-BOT / -MID into the rib notches;
they end at the inner arm panels with tabs through them.
- Slide the 2 ARM-PANEL-INS home over those end tabs.
- Drop RAIL-BACK-TOP in - it runs the full width through both arm panels.
- Thread the 12 BACK-STILEs down through the three back rails.
- Drop the 2 ARM-PANEL-OUTs on, then the 2 ARM-CAPs into the top notches.
- Drop the three SEAT-DECK sectors over the rib tabs.
- Bend and glue the three 4 mm SKINs around the base and the back.
- Glue every joint (PVA D3) and clamp. No fasteners are required.

BILL OF MATERIALS

PART	QTY	SIZE (mm)	FUNCTION
15 mm PLYWOOD			
RIB	5	900 x 730	radial rib, drops onto the base rails
ARM-PANEL-IN	2	900 x 730	closes the seat structure, takes the rail end tabs
ARM-PANEL-OUT	2	900 x 730	the end of the sofa, tied by the arm cap
ARM-CAP	2	231 x 692	arm top plate between both arm panels
RAIL-BASE-IN	3	1433 x 280	3 laminae = 45 mm plinth, floor level
RAIL-BASE-OUT	3	2193 x 382	3 laminae = 45 mm plinth, floor level
RAIL-SEAT-IN	1	1122 x 206	z 315-330 mm, top_notch
RAIL-SEAT-OUT	1	1585 x 254	z 315-330 mm, top_notch
RAIL-BACK-BOT	1	1748 x 280	z 330-345 mm, outer_notch
RAIL-BACK-MID	1	1748 x 280	z 520-535 mm, outer_notch
RAIL-BACK-TOP	1	2203 x 384	z 745-760 mm, post_top_notch
SEAT-DECK-1	1	402 x 697	rests on the ribs and both seat rails
SEAT-DECK-2	1	793 x 716	rests on the ribs and both seat rails
SEAT-DECK-3	1	402 x 697	rests on the ribs and both seat rails
BACK-STILE	12	70 x 430	threads down through the three back rails
4 mm FLEXIBLE PLYWOOD			
SKIN-BASE-OUT	1	2304 x 345	wraps the outer face of the base
SKIN-BASE-IN	1	1361 x 345	wraps the front face of the base
SKIN-BACK	1	1805 x 415	wraps the outer face of the backrest

TOTAL 40 pieces

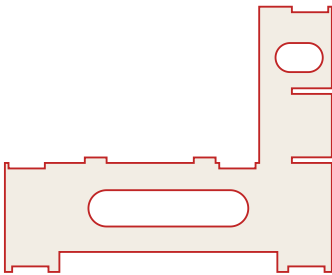


CURVED 3-SEAT SOFA / CNC FRAME

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF

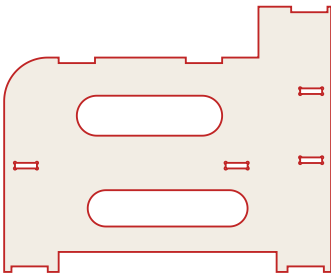
3 - PARTS INDEX - 18 distinct parts, 40 pieces

RIB x5



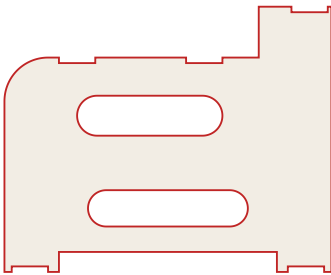
900 x 730 mm
sheet 4,5

ARM-PANEL-IN x2



900 x 730 mm
sheet 6

ARM-PANEL-OUT x2



900 x 730 mm
sheet 6,7

ARM-CAP x2



231 x 692 mm
sheet 3

RAIL-BASE-IN x3



1433 x 280 mm
sheet 3

RAIL-BASE-OUT x3



2193 x 382 mm
sheet 1,2

RAIL-SEAT-IN x1



1122 x 206 mm
sheet 4

RAIL-SEAT-OUT x1



1585 x 254 mm
sheet 3

RAIL-BACK-BOT x1



1748 x 280 mm
sheet 2

RAIL-BACK-MID x1



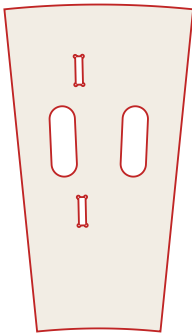
1748 x 280 mm
sheet 2

RAIL-BACK-TOP x1



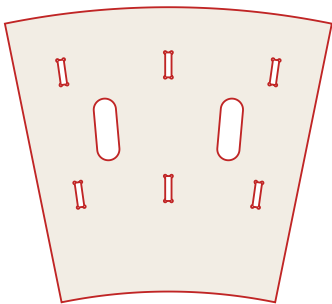
2203 x 384 mm
sheet 1

SEAT-DECK-1 x1



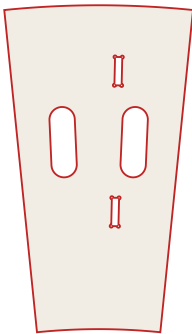
402 x 697 mm
sheet 4

SEAT-DECK-2 x1



793 x 716 mm
sheet 7

SEAT-DECK-3 x1



402 x 697 mm
sheet 7

BACK-STILE x12



70 x 430 mm
sheet 2

SKIN-BASE-OUT x1



2304 x 345 mm
sheet 8

SKIN-BASE-IN x1



1361 x 345 mm
sheet 8

SKIN-BACK x1



1805 x 415 mm
sheet 8

Each part is fitted to its own frame, so the drawings are NOT to a common scale - read the millimetre size under each one.
Blue-tinted parts are 4 mm flexible plywood; everything else is 15 mm. The sheet number tells you which nesting page it is on.
generated 2026-08-24 - curved-sofa.dxf.py - all dimensions in millimetres

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4 - FOAM, FABRIC AND CONSUMABLES

FOAM SCHEDULE

ITEM	SPEC	QTY	PLAN AREA	THICK / HT	VOLUME	NOTE
Seat cushion	HR foam 35 kg/m3	1	0.92 m2	120 mm	110 L	annular sector R1300-1990, 46.2 deg
Back cushion	HR foam 30 kg/m3	1	0.25 m2	415 mm	105 L	415 tall against the lattice, 150 deep
Arm bolster	HR foam 30 kg/m3	2	0.13 m2	220 mm	57 L	over the arm cap; the front roll is carved from the block

TOTAL FOAM 272 litres = 0.272 m3 - about 9 kg, cut from bought blocks against the SEAT-DECK and ARM-CAP parts used as templates

SHEET GOODS

Polyester wadding 200 g/m2	1.68 m2	under every cover
Stretch calico, first cover	1.64 m2	platform + inside back
Black cambric dust cover	1.65 m2	stapled under the plinth

COVER PANEL SCHEDULE

Seat cushion, top and bottom	1.83 m2
Seat cushion border	0.48 m2
Back cushion, front and back	1.41 m2
Back cushion top and ends	0.38 m2
Arm covers, both arms	1.73 m2
Base outer and front faces	1.26 m2
Base ends, both	0.62 m2
Seat platform, inside back	0.92 m2

NET COVER AREA 8.62 m2

FABRIC TO BUY 9 linear metres on a 140 cm bolt

Includes a 50 % allowance: a crescent frame nests badly and every panel is cut on a curve.
Add another 30 % for a directional, striped or patterned fabric.

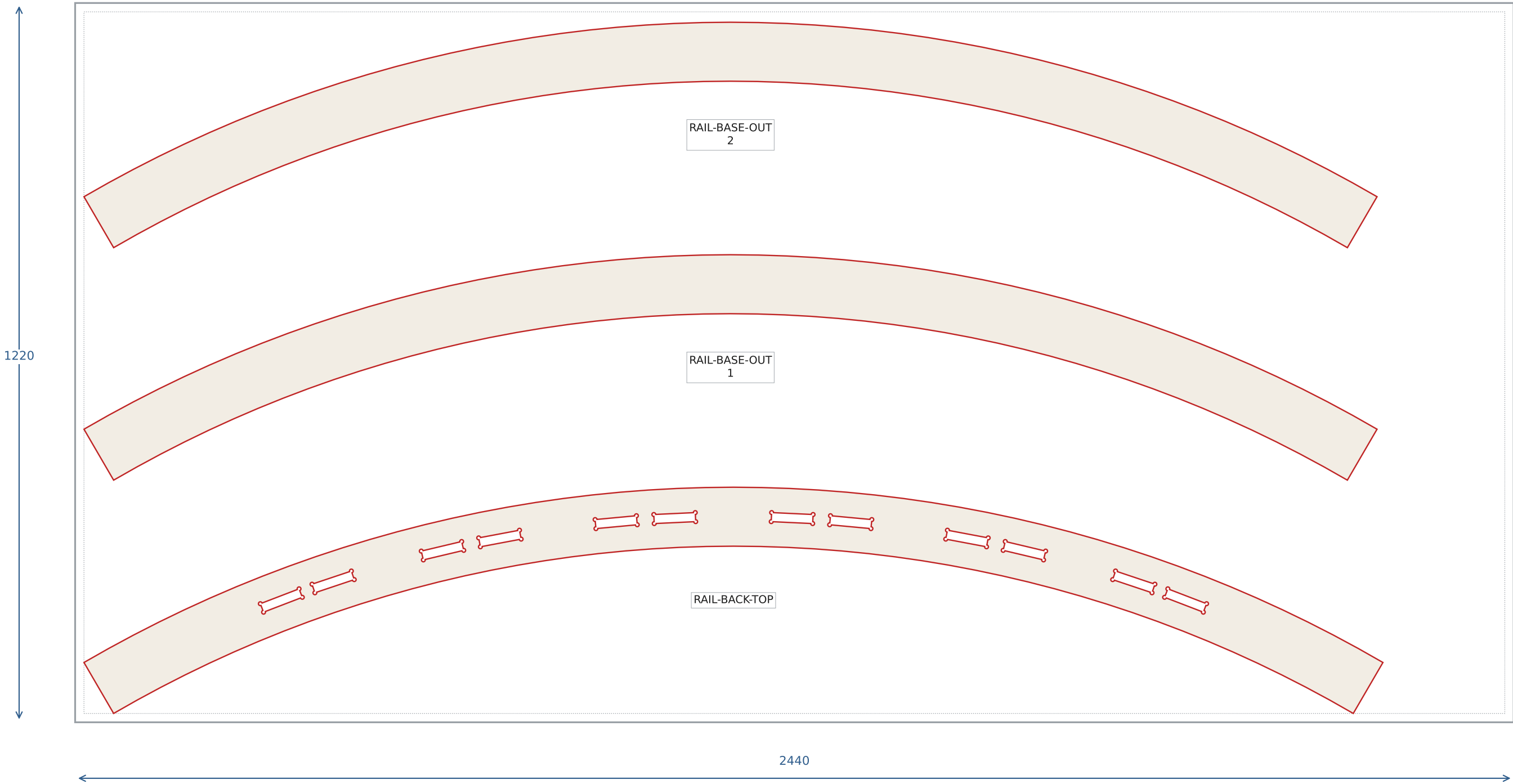
OTHER CONSUMABLES

- Elastic webbing 50 mm: 9 radial runs x 900 mm plus 15 % tension = 9.5 m, with 18 clips.
- Staples 10 mm: roughly 900 for a frame this size.
- PVA D3 wood glue: about 1.0 litre for the plinth lamination and every slot joint.
- Spray contact adhesive for foam: 1 aerosol per 2 m2 of foam face, so 3 cans.
- Feet are not in the cut file - the 45 mm laminated plinth IS the foot. Add bought glides, or a 60 mm metal leg set if the sofa should sit higher.

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5 - NESTING SHEET 1 OF 8 - 15 mm PLYWOOD

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF
RAIL-BACK-TOP x1, RAIL-BASE-OUT x2

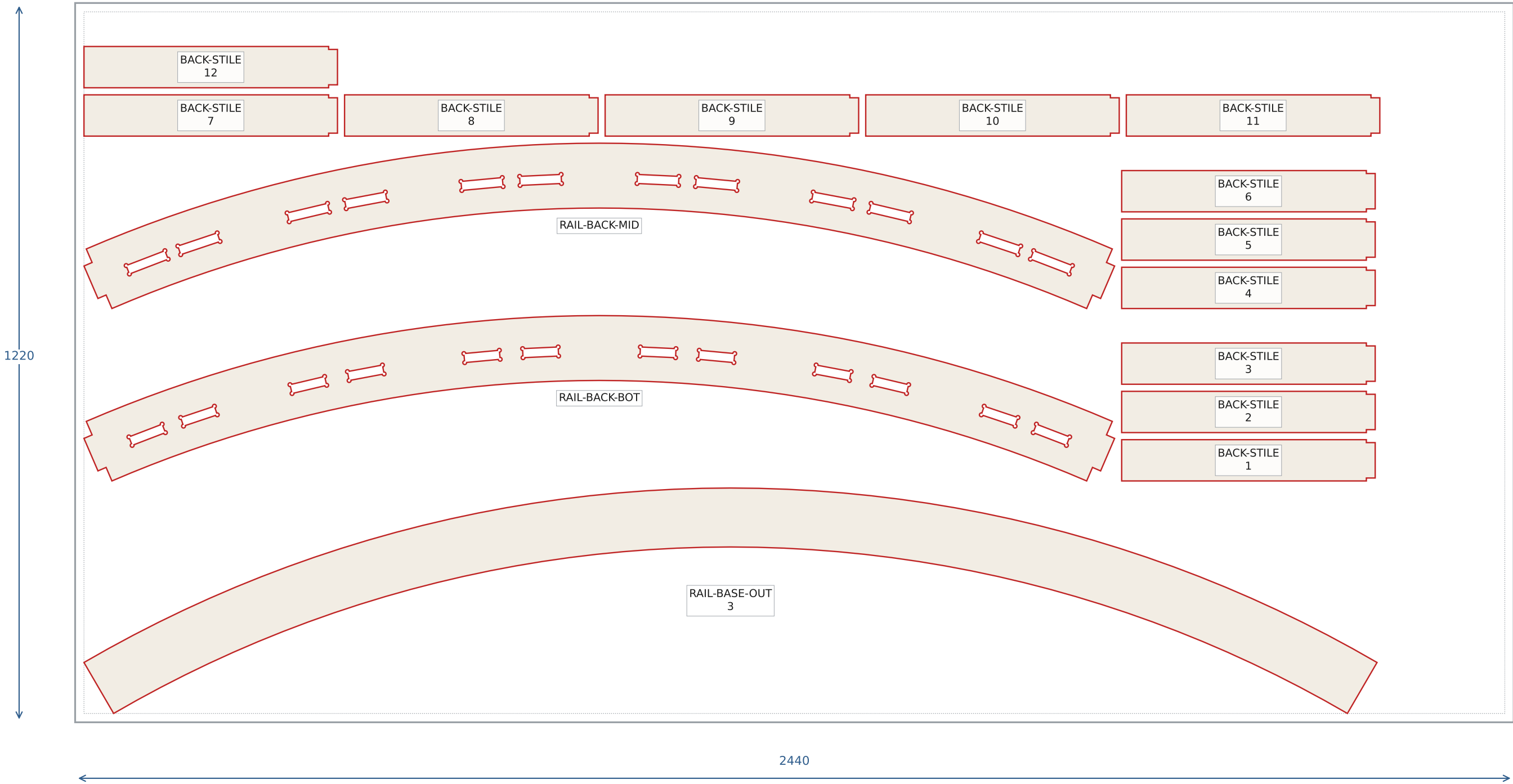


CURVED 3-SEAT SOFA / CNC FRAME

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF

6 - NESTING SHEET 2 OF 8 - 15 mm PLYWOOD

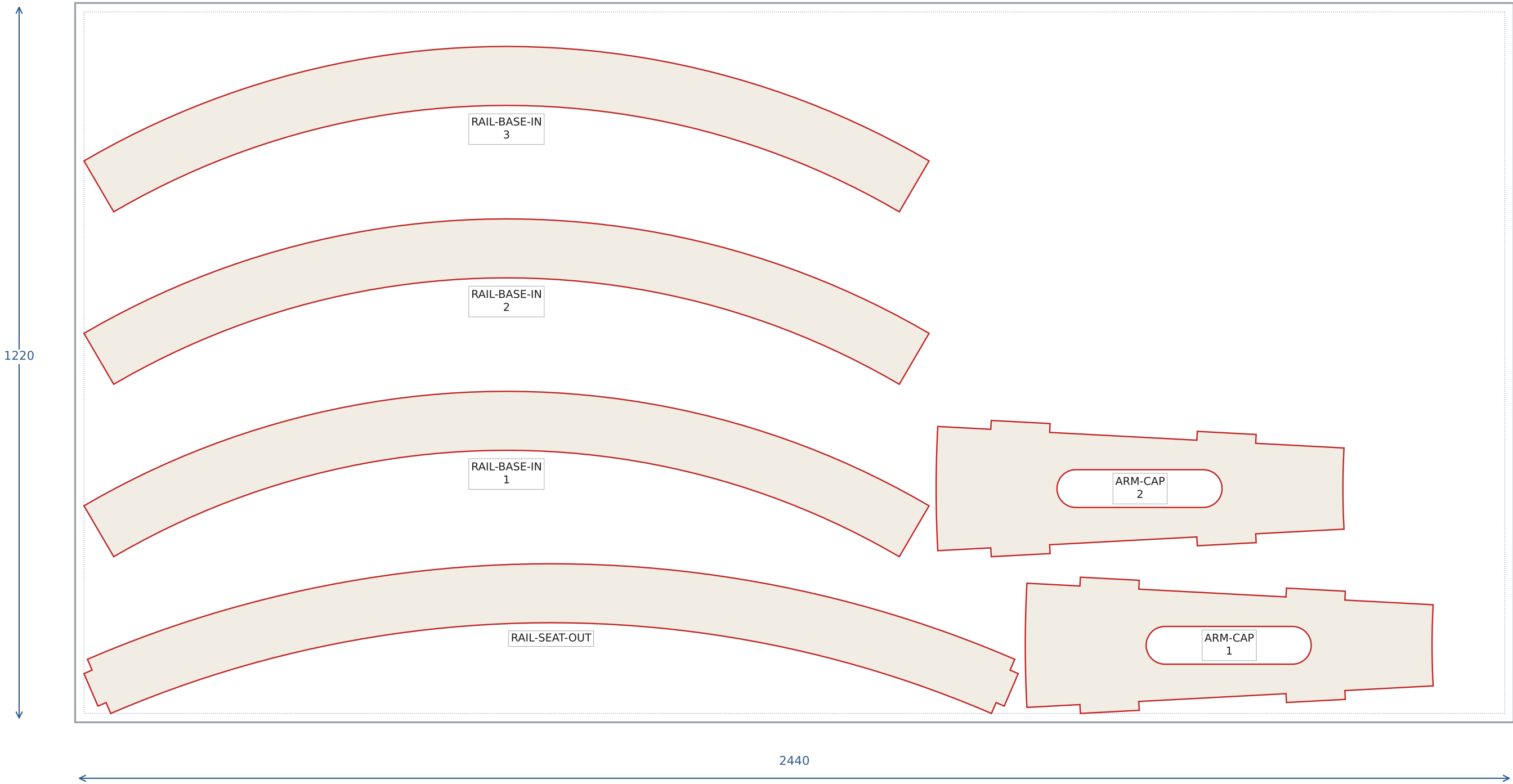
BACK-STILE x12, RAIL-BACK-BOT x1, RAIL-BACK-MID x1, RAIL-BASE-OUT x1



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7 - NESTING SHEET 3 OF 8 - 15 mm PLYWOOD

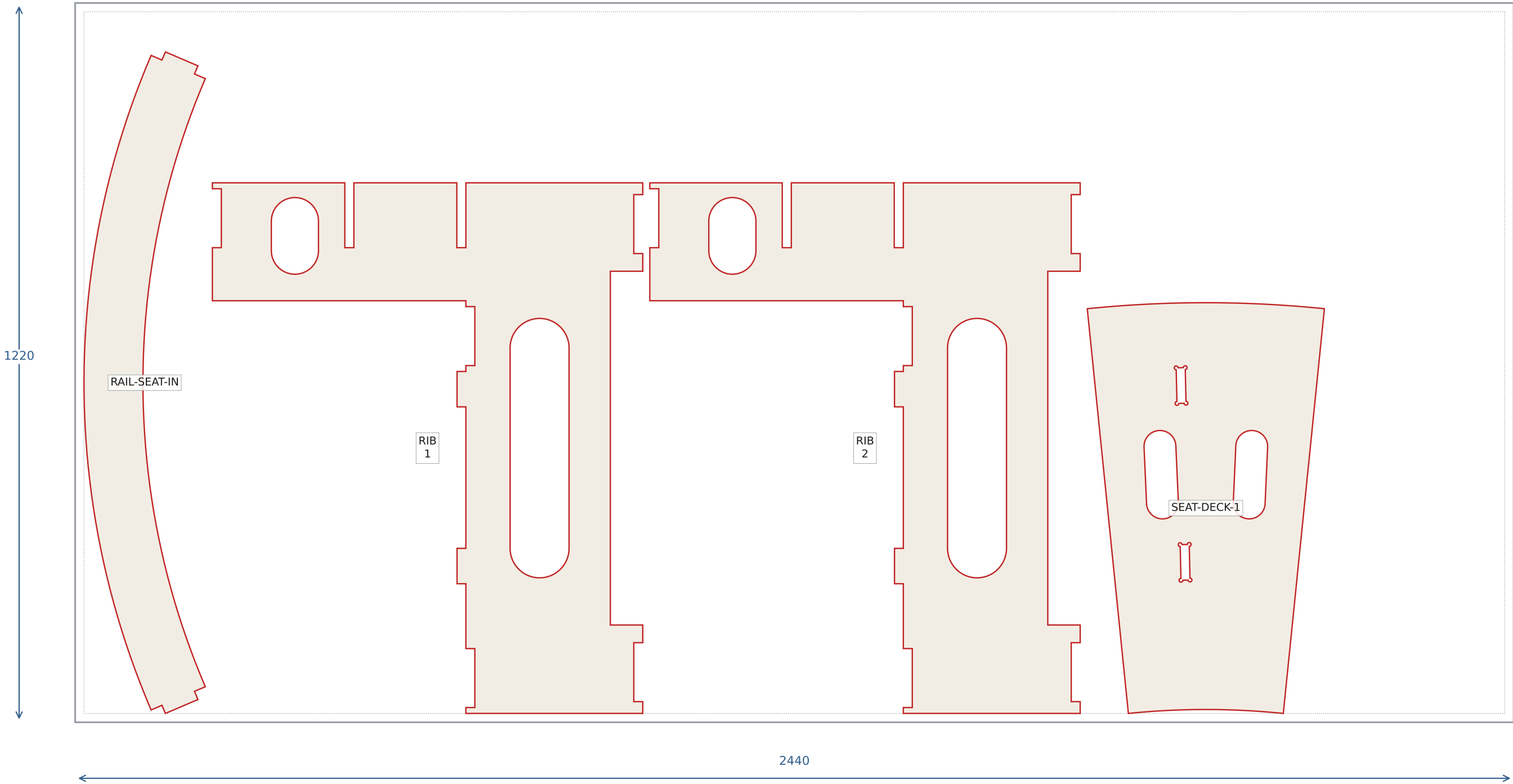
15 mm + 4 mm PLYWOOD | mm | 1:1 DXF
ARM-CAP x2, RAIL-BASE-IN x3, RAIL-SEAT-OUT x1

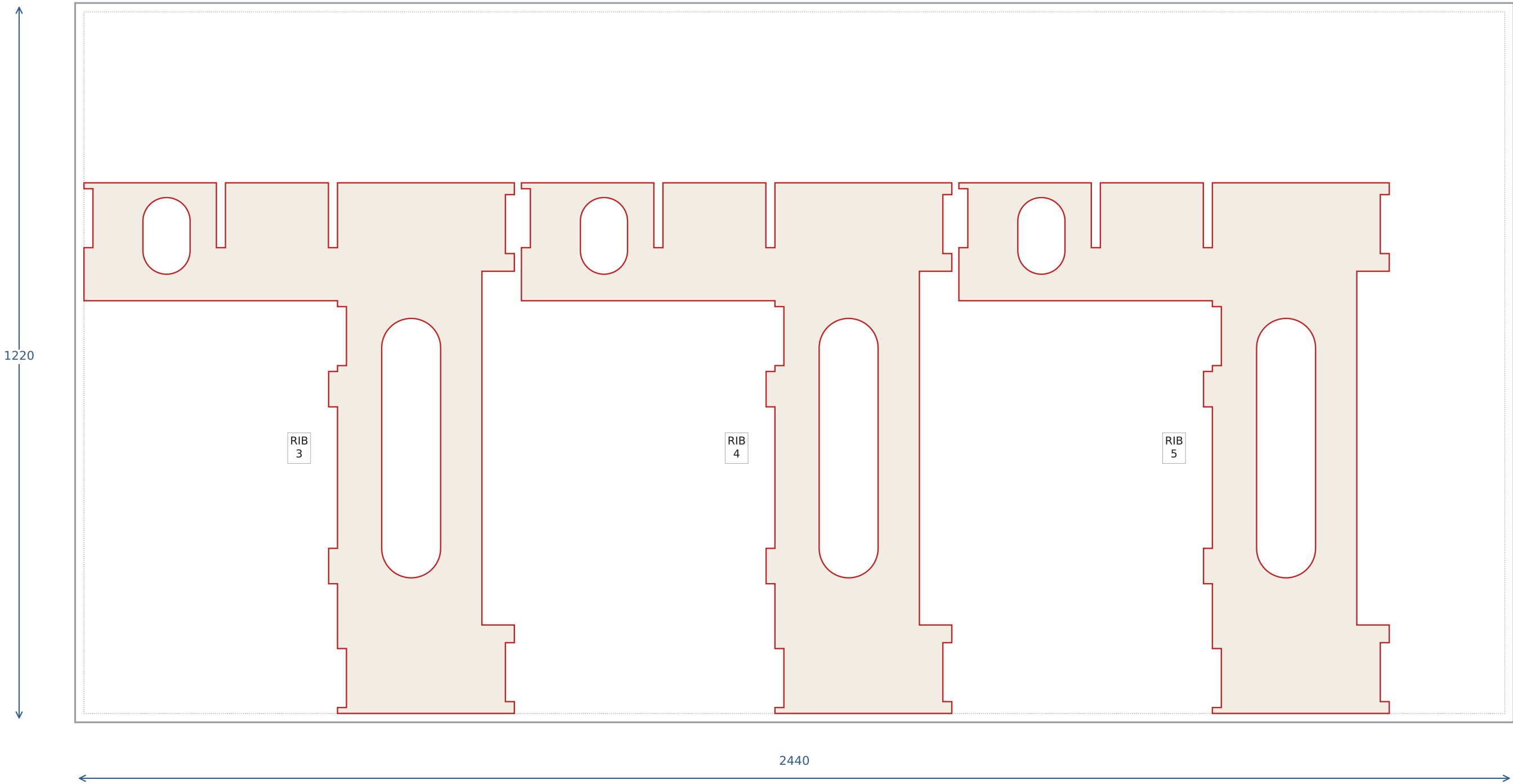


CURVED 3-SEAT SOFA / CNC FRAME

8 - NESTING SHEET 4 OF 8 - 15 mm PLYWOOD

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF
RAIL-SEAT-IN x1, RIB x2, SEAT-DECK-1 x1

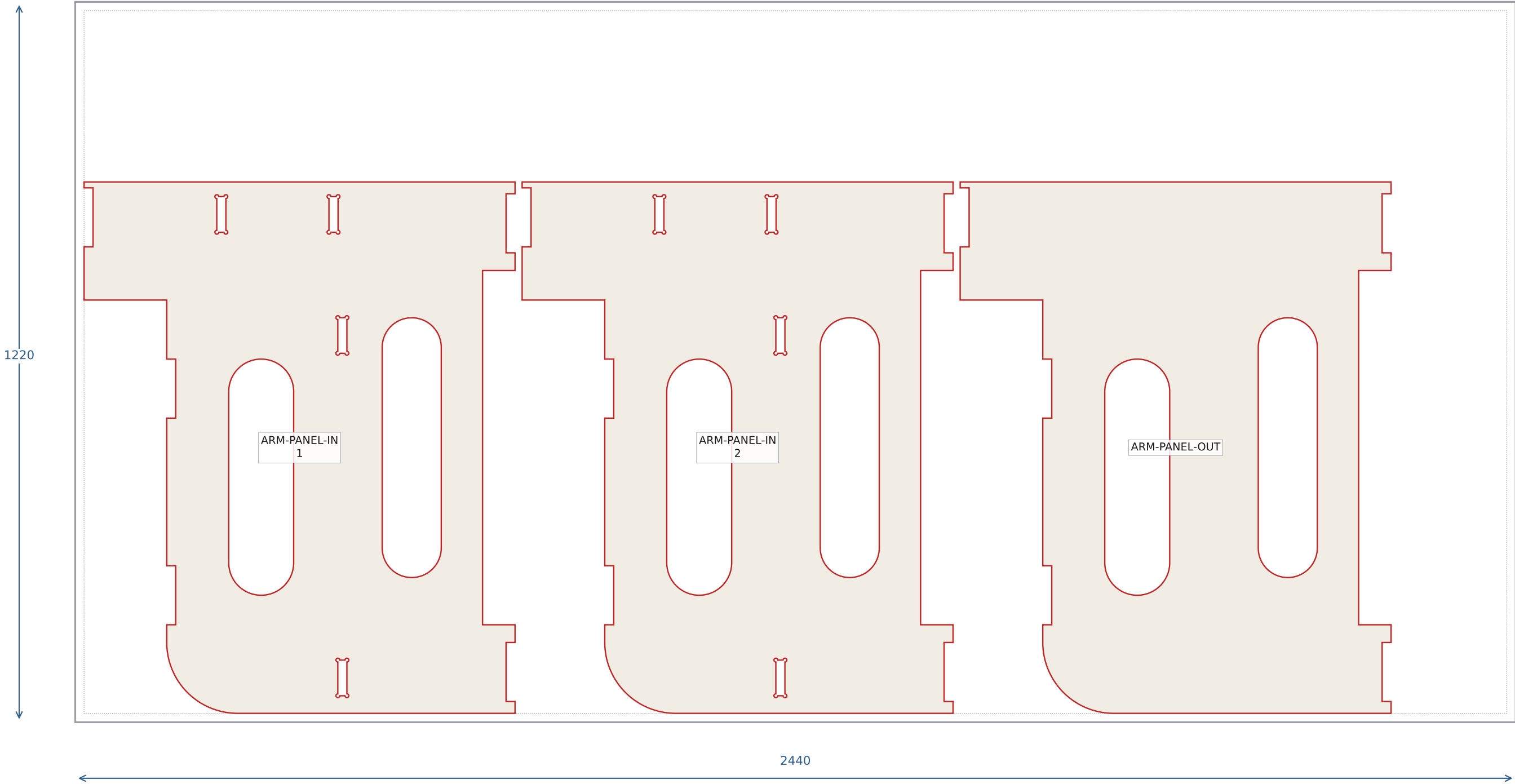




CURVED 3-SEAT SOFA / CNC FRAME

10 - NESTING SHEET 6 OF 8 - 15 mm PLYWOOD

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF
ARM-PANEL-IN x2, ARM-PANEL-OUT x1

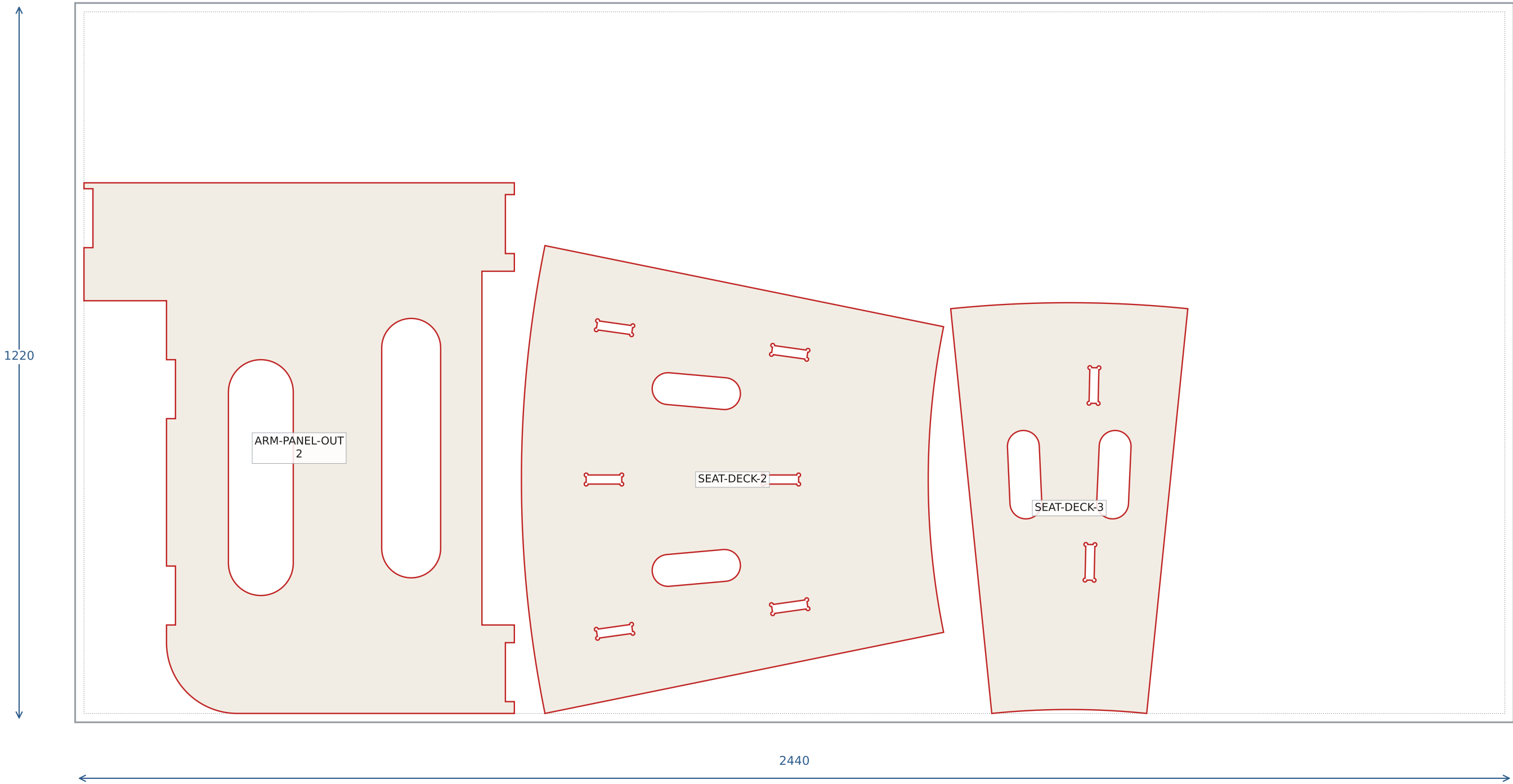


CURVED 3-SEAT SOFA / CNC FRAME

11 - NESTING SHEET 7 OF 8 - 15 mm PLYWOOD

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF

ARM-PANEL-OUT x1, SEAT-DECK-2 x1, SEAT-DECK-3 x1



CURVED 3-SEAT SOFA / CNC FRAME

12 - NESTING SHEET 8 OF 8 - 4 mm FLEXIBLE PLYWOOD

15 mm + 4 mm PLYWOOD | mm | 1:1 DXF

SKIN-BACK x1, SKIN-BASE-IN x1, SKIN-BASE-OUT x1

